

Shijun ZHANG

Assistant Professor

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Appointments

Assistant Professor (Tenure-Track), Hong Kong Polytechnic University (PolyU), Hong Kong SAR, China	Jul 2024 ~ Present
Phillip Griffiths Assistant Research Professor , Duke University, United States Mentors: Jianfeng Lu and Hongkai Zhao	Aug 2022 ~ Jun 2024
Research Fellow , National University of Singapore, Singapore Mentor: Zuowei Shen	Jan 2021 ~ Jul 2022

Education

Ph.D. in Mathematics , National University of Singapore, Singapore Thesis: <i>Deep neural network approximation via function compositions</i> [PDF, URL] Supervisors: Zuowei Shen and Haizhao Yang	Aug 2016 ~ Jan 2021
B.S. in Mathematics , Wuhan University, China Thesis supervisor: Xiliang Lv	Sep 2012 ~ Jun 2016

Awards

The 2026 Frontiers of Science Award (Mathematics), International Congress for Basic Science (ICBS 2026), URL.

Departmental Best Paper Award, AMA, PolyU, 2024-2025, URL.

The EASIAM (East Asia section of SIAM) Student Paper Prize (First Prize), 2020-2021, URL.

Teaching

Operations Research Methods (AMA 3820), PolyU Instructor, Syllabus	2025/26 Sem1, 2025/26 Sem2
Optimization Methods (AMA 4850), PolyU Instructor, Syllabus	2024/25 Sem2, 2025/26 Sem2
Mathematical Numerical Analysis (Math 361S), Duke University Instructor, Syllabus	Spring 2024
Matrices and Vectors (Math 218D-2), Duke University Teaching assistant	Fall 2023

Publications

[number] Author(s). *Paper title*. Journal or conference reference. [Links]

* Corresponding author † Equal contribution

Preprint(s)

- [21] Baicheng Li, Haizhao Yang, **Shijun Zhang**. *Sobolev Approximation by Fixed-Size Neural Networks with Arbitrary Accuracy*. Submitted. [arXiv]
- [20] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Geometric Layer-wise Approximation Rates for Deep Networks*. Submitted. [arXiv]
- [19] Zeyu Li, **Shijun Zhang**, Tiejong Zeng, Feng-Lei Fan. *Neural Network Approximation: A View from Polytope Decomposition*. Submitted. [arXiv]
- [18] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Multigrade neural network approximation*. Submitted. [arXiv]
- [17] Hanyu Pei, Jing-Xiao Liao, Qibin Zhao, Ting Gao, **Shijun Zhang**, Xiaoge Zhang, Feng-Lei Fan. *NeuronSeek: On Stability and Expressivity of Task-driven Neurons* Submitted. [arXiv]

Published (Accepted)

- [16] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Fourier Multi-Component and Multi-Layer Neural Networks: Unlocking High-Frequency Potential*. Accepted by Neural Networks. [arXiv]
- [15] Fenglei Fan, Juntong Fan, Dayang Wang, Jingbo Zhang, Zelin Dong, **Shijun Zhang**, Ge Wang, Tiejong Zeng. *Hyper-Compression: Model Compression via Hyperfunction*. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), DOI: 10.1109/TPAMI.2026.3673772, to appear. [arXiv, Journal]
- [14] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Structured and balanced multi-component and multi-layer neural networks*. **SIAM Journal on Scientific Computing**, 47(5):C1059-C1090, 2025. [arXiv, Journal]
- [13] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Why shallow networks struggle to approximate and learn high frequencies*. **Information and Inference: A journal of the IMA**, Volume 14, Issue 3, September 2025, iaaf022, <https://doi.org/10.1093/imaiai/iaaf022>. [arXiv, Journal]
- [12] Qianchao Wang[†], **Shijun Zhang**[†], Dong Zeng, Zhaocheng Xie, Hengtao Guo, Feng-Lei Fan, Tiejong Zeng. *Don't Fear Peculiar Activation Functions: EUAF and Beyond*. **Neural Networks**, 186, June 2025. [arXiv, Journal]
- [11] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *Deep network approximation: Beyond ReLU to diverse activation functions*. **Journal of Machine Learning Research**, 25(35):1–39, 2024. [arXiv, Journal]
- [10] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *On enhancing expressive power via compositions of single fixed-size ReLU network*. Proceedings of the 40th International Conference on Machine Learning (ICML 2023), PMLR 202:41452–41487, 2023. [arXiv, Poster, Conference]
- [9] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Neural network architecture beyond width and depth*. Advances in Neural Information Processing Systems (NeurIPS 2022), 35:5669–5681, 2022. [arXiv, Poster, Conference]

- [8] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation in terms of intrinsic parameters*. Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**), PMLR 162:19909–19934, 2022. [arXiv, Spotlight, Conference]
- [7] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation: achieving arbitrary accuracy with fixed number of neurons*. **Journal of Machine Learning Research**, Volume 23, Issue 276, September 2022, Pages 1–60. [arXiv, Journal]
- [6] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Optimal approximation rate of ReLU networks in terms of width and depth*. **Journal de Mathématiques Pures et Appliquées**, Volume 157, January 2022, Pages 101–135. [arXiv, Journal]
- [5] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Neural network approximation: Three hidden layers are enough*. **Neural Networks**, Volume 141, September 2021, Pages 160–173. [arXiv, Journal]
- [4] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network with approximation error being reciprocal of width to power of square root of depth*. **Neural Computation**, Volume 33, Issue 4, April 2021, Pages 1005–1036. [arXiv, Journal]
- [3] Jianfeng Lu, Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation for smooth functions*. **SIAM Journal on Mathematical Analysis**, Volume 53, Issue 5, September 2021, Pages 5465–5506. [arXiv, Journal]
- [2] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation characterized by number of neurons*. **Communications in Computational Physics**, Volume 28, Issue 5, November 2020, Pages 1768–1811. [arXiv, Journal]
- [1] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Nonlinear approximation via compositions*. **Neural Networks**, Volume 119, November 2019, Pages 74–84. [arXiv, Journal]